Algebra 1
Unit 8
Lesson 9 Homework

Name Answer Key
Date____________________
Period____________________

A theater owner is looking to maximize his profits and needs to increase his ticket prices. The function $P(t) = -50t^2 + 100t + 14{,}400$ represents his total profit, P, based on the amount, in dollars, he raises the ticket price, t.

1. Rewrite the function in vertex form.

$$P(t) = -50(t-1)^2 + 14{,}450$$

2. Rewrite the function in factored form.

$$P(t) = -50(t-18)(t+16)$$

Complete the statements.

3. The
   - ○ vertex
   - ○ factored
   - ○ standard

   form of the function can be used to determine the interval on which the theater owner experiences a profit. The theater own experiences a profit on the interval ________$0 \le t \le 18$____________.

4. The
   - ○ vertex
   - ○ factored
   - ○ standard

   form of the function can be used to determine when the theater owner experiences the greatest profit. The theater owner experiences the greatest profit when __________$t = 1$___________.

5. The
   - ○ vertex
   - ○ factored
   - ○ standard

   form of the function can be used to determine when the theater owner experiences neither a profit nor a loss. This occurs when the ticket price increased by $18.

The function $H(x) = 0.5x^2 - x + 33$ models the amount of people, in millions, H, enrolled in an HMO insurance in the United States in the years, x, since 1990.

6. What would the zeros of this function reveal in this context?

The number of years since 1990 when there would be 0 people enrolled in the plan. Sincer there are no zeros, there are no years with 0 people enrolled.

7. Rewrite the function in vertex form.

$$H(x) = 0.5(x - 1)^2 + 32.5$$

Complete the statements.

8. The
   - vertex
   - factored
   - standard

   form of the function can be used to determine the year with the lowest HMO enrollment.

9. The year with the lowest HMO enrollment was _1991_________ when _32.5_____ million people were enrolled.

10. The
    - vertex
    - factored
    - standard

    form of the function can be used to determine how many people, in millions, were enrolled in an HMO in 1990, ____33_____ million people.